Vasopressin: Drug information - UpToDate

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Contributor Disclosures

For additional information see ”Vasopressin: Patient drug information” and ”Vasopressin: Pediatric drug information”

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Vasostrict

Pharmacologic Category

- Antidiuretic Hormone Analog;
- Hormone, Posterior Pituitary

Dosing: Adult

Arginine vasopressin deficiency, acute

Arginine vasopressin deficiency, acute (formerly called central diabetes insipidus) (off-label use): Note: May be considered for short-term use following transsphenoidal surgery in the neurocritical care setting. Use is generally limited to the early postoperative period (eg, days 1 to 4) since arginine vasopressin deficiency (AVP-D) may spontaneously resolve or revert to syndrome of inappropriate antidiuretic hormone secretion (Ref). To avoid hyponatremia, use the minimum effective dose to control polyuria (Ref).

Continuous infusion: **IV:** Reported dosing ranges vary widely; refer to institutional protocols. Monitor urine output frequently and adjust infusion rate accordingly (Ref).

SUBQ (off-label route): 5 to 10 units 2 to 3 times daily as needed (Ref).

Deceased organ donor management

Deceased organ donor management (hormone replacement therapy) (off-label use): Note: Use if hypotensive despite adequate fluid resuscitation, left ventricular ejection fraction <45%, low systemic vascular resistance (SVR), or if AVP-D is present. Use as part of combination hormone therapy (Ref).

IV: Initial: 0.01 to 0.04 units/minute (Ref); some experts recommend a bolus of 1 unit followed by 0.01 to 0.1 units/minute titrated to a SVR of 800 to 1,200 dyne•second/cm^5 (usual dose: 0.01 to 0.04 units/minute) (Ref).

Septic shock and other vasodilatory shock states

Septic shock and other vasodilatory shock states (adjunctive agent): Note: Consider in patients requiring escalating doses of norepinephrine (eg, 0.2 to 0.5 mcg/**kg**/min) (Ref).

Shock, sepsis:

Continuous infusion: **IV:** 0.03 units/minute added to norepinephrine as a fixed dose (not titrated); usual dosage range: 0.01 to 0.04 units/minute; doses >0.04 units/minute should be reserved for salvage therapy due to potential risk of ischemic complications (eg, skin, cardiac, splanchnic) (Ref).

Discontinuation in septic shock: Multiple studies describe clinically significant hypotension when vasopressin is discontinued prior to norepinephrine. When discontinuing therapy, consider slowly tapering by 0.01 units/minute every 30 to 60 minutes to reduce the risk of hypotension; monitor MAP when discontinuing vasopressin (Ref).

Shock, post-cardiotomy: Continuous infusion: **IV:** Initial: 0.03 units/minute. If the target blood pressure response is not achieved, titrate up by 0.005 units/minute at 10- to 15-minute intervals (maximum dose: 0.1 units/minute). After target blood pressure has been maintained for 8 hours without the use of catecholamines, taper by 0.005 units/minute every hour as tolerated to maintain target blood pressure.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: IV, SUBQ: No dosage adjustment necessary for any degree of kidney impairment (limited urinary excretion of unchanged drug) (Ref).

Hemodialysis, intermittent (thrice weekly): Unlikely to be significantly dialyzable (Ref): **IV, SUBQ:** No supplemental dose or dosage adjustment necessary (Ref).

Peritoneal dialysis: Unlikely to be significantly dialyzable: **IV, SUBQ:** No dosage adjustment necessary (Ref).

CRRT: Some removal by CRRT but clinical significance is unknown (Ref): **IV, SUBQ:** No dosage adjustment necessary (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): Dialyzability unknown: **IV, SUBQ:** No dosage adjustment necessary (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Vasopressin: Pediatric drug information")

Note: Units of measure vary by indication and age (ie, milliunits/kg/**hour**, units/kg/**hour**, milliunits/kg/minute, units/kg/minute; **units/minute**, units/**hour**); extra precautions should be taken.

Arginine vasopressin deficiency

Arginine vasopressin deficiency (AVP-D) (formerly called central diabetes insipidus): Limited data available:

Note: Highly variable dosage; titrate dosage based upon serum and urine sodium and osmolality in addition to fluid balance and urine output.

Continuous IV infusion:

Infants, Children, and Adolescents: Continuous IV infusion: Initial: 0.5 milliunits/kg/**hour**; titrate upward in 0.5 milliunits/kg/**hour** increments at approximately 10-minute intervals to target urine output (suggested output target: <2 mL/kg/hour) (Ref); infusion rates up to 10 milliunits/kg/**hour** have been reported (Ref). **Note:** Use in conjunction with fluid therapy, monitor urine output and specific gravity, serum and urine electrolytes (primarily Na), and plasma osmolality.

IM, SUBQ:

Children and Adolescents: IM, SUBQ: 2.5 to 10 units 2 to 4 times daily (Ref).

Cadaveric organ donations

Cadaveric organ donations (hormone replacement therapy): Limited data available (Ref): Infants, Children, and Adolescents: Continuous IV infusion: Initial: 0.5 to 1 milliunits/kg/**hour**; titrate dose to maintain targeted urine output. In a retrospective case-controlled study (n=34, age: 8.2 ± 6.2 years), a mean dose of 41 milliunits/kg/**hour** was reported with a range of 0.2 milliunits/kg/**hour** to 150 milliunits/kg/**hour** (Ref).

GI hemorrhage

GI hemorrhage: Limited data available: Children and Adolescents: Continuous IV infusion: Initial: 2 to 5 milliunits/kg/minute; titrate dose as needed; maximum dose: 10 milliunits/kg/minute (Ref); usual adult dosing range is 0.2 to 0.4 **units/minute** up to a maximum rate of 0.8 **units/minute** (Ref). **Note:** Higher doses have not been associated with greater efficacy but have been associated with increased risk of complications (Ref).

Pulseless arrest, ventricular fibrillation, ventricular tachycardia

Pulseless arrest, ventricular fibrillation, ventricular tachycardia: Limited data available: Infants, Children, and Adolescents: IV: 0.4 units/kg as a single dose after traditional resuscitation methods and at least two doses of epinephrine have been administered. **Note:** Due to insufficient evidence, no formal recommendations for or against the routine use of vasopressin during pediatric cardiac arrest are provided (Ref).

Vasodilatory shock with hypotension unresponsive to fluid resuscitation and exogenous catecholamines

Vasodilatory shock with hypotension unresponsive to fluid resuscitation and exogenous catecholamines: Limited data available; efficacy results have varied (Ref): Infants, Children, and Adolescents: Continuous IV infusion: 0.17 to 8 milliunits/kg/minute (0.01 to 0.48 units/kg/**hour**) has been used; dosing based on retrospective reviews and case reports that have shown increases in arterial blood pressure and urine output and also allowed for dosage reductions of additional vasopressors (Ref). The only double-blind, placebo-controlled trial (n=65, age: 3 to 14 years) evaluated a dose of 0.5 to 2 milliunits/kg/minute (0.03 to 0.12 units/kg/**hour**) at multiple centers and found no significant difference in the time to reach hemodynamic stability and a trend toward increased mortality in the vasopressin group was noted. Based on these findings, the authors did not recommend routine use of vasopressin (Ref). **Note:** Abrupt discontinuation of infusion may result in hypotension; to discontinue, gradually taper infusion.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified.

Frequency not defined:

Cardiovascular: Atrial fibrillation, bradycardia, ischemic heart disease, limb ischemia (distal), low cardiac output, right heart failure, shock (hemorrhagic)

Dermatologic: Skin lesion (ischemic)

Endocrine & metabolic: Hyponatremia

Gastrointestinal: Mesenteric ischemia

Hematologic & oncologic: Decreased platelet count, hemorrhage (intractable)

Hepatic: Increased serum bilirubin

Renal: Renal insufficiency

Contraindications

Hypersensitivity to vasopressin or any component of the formulation; hypersensitivity to chlorobutanol (chlorobutanol-containing formulations only).

Canadian labeling: Additional contraindications (not in US labeling): Cardiorenal disease with hypertension; advanced arteriosclerosis; coronary thrombosis; angina; chronic nephritis with nitrogen retention; epilepsy or toxemia of pregnancy.

Warnings/Precautions

Concerns related to adverse effects:

• Arginine vasopressin disorders: Reversible arginine vasopressin disorders (formerly called diabetes insipidus) may occur following discontinuation of vasopressin therapy; signs/symptoms may include polyuria, dilute urine, and hypernatremia. Monitor electrolytes, fluid status, and urine output after vasopressin discontinuation. Readministration of vasopressin (or desmopressin) may be needed to correct fluid and electrolyte imbalances.

• Extravasation: Vesicant; ensure proper needle or catheter placement prior to and during infusion. Extravasation may lead to severe vasoconstriction and localized tissue necrosis; also, gangrene of extremities, tongue, and ischemic colitis. Avoid extravasation.

• Water intoxication: May cause water intoxication; early signs include drowsiness, listlessness, and headache; these should be recognized to prevent coma and seizures.

Disease-related concerns:

• Cardiovascular disease: Use with caution in patients with cardiovascular disease, including arteriosclerosis; may worsen cardiac output.

• Migraine: Use with caution in patients with a history of migraines.

• Renal impairment: Use with caution in patients with renal disease.

• Vascular disease: Use with caution in patients with vascular disease.

Warnings: Additional Pediatric Considerations

May cause hyponatremia; frequency not well-defined in pediatric patients and may vary based on indication. In young pediatric patients (mean age: 5.2 months; including neonates) who received a vasopressin infusion for hemodynamic instability following complex cardiac surgery, the observed incidence of hyponatremia (defined as serum Na <135 mEq/L) was statistically higher in the vasopressin group than the control (48% vs 17%, p=0.004) and the decrease in serum sodium concentration was significantly greater (9.9 mEq/L vs 7.5 mEq/L, p=0.009); it was also observed that the decrease in the serum Na was more rapid in the vasopressin group (Davalos 2013). With other uses in pediatric patients, hyponatremia has not been characterized; monitor fluid balance, serum Na and urine output closely.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Vasostrict: 20 units/mL (1 mL)

Vasostrict: 20 units/mL (10 mL) [contains chlorobutanol (chlorobutol)]

Generic: 20 units/mL (1 mL, 10 mL); 20 units/100 mL in NaCl 0.9% (100 mL); 40 units/100 mL in NaCl 0.9% (100 mL)

Solution, Intravenous [preservative free]:

Vasostrict: 20 units/100 mL in Dextrose 5% (100 mL); 40 units/100 mL in Dextrose 5% (100 mL)

Generic: 20 units/mL (1 mL); 20 units/100 mL in NaCl 0.9% (100 mL); 40 units/100 mL in NaCl 0.9% (100 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Vasopressin Intravenous)

20 units/mL (per mL): $21.61 - $126.13

Solution (Vasopressin-Sodium Chloride Intravenous)

20UT/100ML 0.9% (per mL): $0.78 - $0.90

40UT/100ML 0.9% (per mL): $1.56 - $1.80

Solution (Vasostrict Intravenous)

20 units/mL (per mL): $11.90

20UT/100ML 5% (per mL): $0.41

40UT/100ML 5% (per mL): $0.82

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection:

Generic: 20 units/mL (1 mL, 2 mL, 5 mL)

Administration: Adult

SUBQ (off-label route): Administer without further dilution.

Continuous IV infusion: Dilute 1 mL and 10 mL vials prior to administration; premixed bags and vials (20 units/100 mL and 40 units/100 mL) requiring no further dilution are also available for single use. Infusion through central line is recommended.

Vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation.

Extravasation management: If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula; elevate extremity; apply dry, warm compresses. Initiate topical nitroglycerin (Ref).

Nitroglycerin (topical): Nitroglycerin topical 2% ointment: Apply a 1-inch strip to the site of ischemia to cover the affected area; may repeat every 8 hours as necessary (Ref).

Alternatives to nitroglycerin (Ref):

Phentolamine: **SUBQ:** Dilute 5 to 10 mg in 10 mL 0.9% sodium chloride and administer into extravasation site as soon as possible after extravasation; if infiltrated catheter has not been removed, administer initial dose through the infiltrated catheter; may repeat in 60 minutes if patient remains symptomatic (Ref).

Terbutaline:

Large extravasations: **SUBQ:** Infiltrate affected extravasation area with terbutaline 1 mg using a solution of terbutaline 1 mg diluted in 10 mL 0.9% sodium chloride; may repeat dose after 15 minutes (Ref).

Small extravasations: **SUBQ:** Infiltrate affected extravasation area with terbutaline 0.5 mg using a solution of terbutaline 1 mg diluted in 1 mL 0.9% sodium chloride; may repeat dose after 15 minutes (Ref).

Preparation for Administration: Adult

Note: Premixed bags and vials (20 units/100 mL and 40 units/100 mL) requiring no further dilution are also available for single use.

For IV administration:

No fluid restriction:

Final concentration 0.1 units/mL: Dilute vasopressin 50 units (2.5 mL) with 500 mL NS or D5W.

Final concentration 0.2 units/mL: Dilute vasopressin 20 units (1 mL) with 100 mL D5W (Ref).

Final concentration 0.4 units/mL: Dilute vasopressin 40 units (2 mL) with 100 mL NS (Ref).

Fluid restriction: Final concentration 1 unit/mL: Dilute vasopressin 100 units (5 mL) with 100 mL NS or D5W.

Usual Infusion Concentrations: Adult

IV infusion: 50 units in 500 mL (concentration: 0.1 unit/mL) **or** 100 units in 100 mL (concentration: 1 unit/mL) of D_5W or NS

Administration: Pediatric

Parenteral:

IM, SUBQ: Administer without further dilution.

Continuous IV infusion: Dilute prior to administration. Infusion through central line is strongly recommended.

Vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation. If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula unless needed for IV antidote; elevate extremity; apply dry, warm compresses. Initiate phentolamine (or alternative antidote) (Ref).

Preparation for Administration: Pediatric

Note: Premixed bags and vials (0.2 units/mL and 0.4 units/mL) requiring no further dilution are also available for single use; some dosage forms may contain preservatives.

Parenteral: Continuous IV infusion: Dilute in NS or D5W to a final concentration of 0.1 to 1 unit/mL. Use of a more dilute solution (0.04 units/mL) was described in a pediatric clinical trial that utilized lower dosing (eg, 0.0005 units/kg/**hour**) (Ref). ASHP recommends standard concentrations of 0.04 units/mL, 0.2 units/mL, or 1 unit/mL in pediatric patients (Ref).

Usual Infusion Concentrations: Pediatric

IV infusion: 0.04 units/mL, 0.2 units/mL, or 1 unit/mL.

Vesicant/Extravasation Risk

Vesicant

Storage/Stability

Premixed bags (not requiring dilution): Store in light-protective carton between 2°C and 8°C (36°F and 46°F). Do not freeze. May also store at ≤25°C (≤77°F) in the carton to protect from light for up to 6 months or manufacturer expiration date, whichever is earlier. Once stored at room temperature, do not return bags to the refrigerator.

Premixed vials (not requiring dilution): Store intact vials in light-protective carton between 2°C and 8°C (36°F and 46°F). Do not freeze. May also remove intact vials in light-protective carton from refrigeration and store at up to 25°C (77°F) for up to 3 to 12 months (varies by manufacturer; refer to product labeling) or manufacturer expiration date, whichever is earlier. Once stored at room temperature, do not return vials to the refrigerator.

Vials (requiring dilution): Store intact vials between 2°C and 8°C (36°F and 46°F). Do not freeze. May also remove intact vials from refrigeration and store at 20°C to 25°C (68°F to 77°F) for up to 12 months or manufacturer expiration date, whichever is earlier (indicate date of removal on the vial or note the manufacturer expiration date, whichever is earlier). After initial entry into the 10 mL vial, opened vial must be refrigerated; discard 30 days after first puncture. Discard unused diluted solution (in D5W or NS) after 18 hours at room temperature or 24 hours under refrigeration.

Use: Labeled Indications

Septic shock and other vasodilatory shock states: To increase BP in adults with vasodilatory shock (eg, postcardiotomy or sepsis) who remain hypotensive despite fluid resuscitation and catecholamines.

Use: Off-Label: Adult

Arginine vasopressin deficiency, acute (formerly called central diabetes insipidus); Deceased organ donor management (hormone replacement therapy)

Medication Safety Issues

Sound-alike/look-alike issues:

PIT or Pitressin (old names sometimes used to describe vasopressin) may be confused with Pitocin

Vasopressin may be confused with desmopressin

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication (IV or intraosseous administration) among its list of drugs which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Administration issues:

Use care when prescribing and/or administering vasopressin solutions. Close attention should be given to concentration of solution, route of administration, dose, and rate of administration (units/minute, units/kg/minute, units/kg/hour).

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Alpha-/Beta-Agonists (Direct-Acting): May increase hypertensive effects of Vasopressin. The effect of other hemodynamic parameters may also be enhanced. *Risk C: Monitor*

Baxdrostat: Hyponatremia-Associated Agents may increase hyponatremic effects of Baxdrostat. *Risk C: Monitor*

Bradycardia-Causing Agents: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Ceritinib: Bradycardia-Causing Agents may increase bradycardic effects of Ceritinib. Management: If this combination cannot be avoided, monitor patients for evidence of symptomatic bradycardia, and closely monitor blood pressure and heart rate during therapy. *Risk D: Consider Therapy Modification*

Desmopressin: Hyponatremia-Associated Agents may increase hyponatremic effects of Desmopressin. *Risk C: Monitor*

Drugs Suspected of Causing Diabetes Insipidus: May decrease therapeutic effects of Vasopressin. Specifically, the pressor and antidiuretic hormone effects of vasopressin may be decreased. *Risk C: Monitor*

Drugs Suspected of Causing SIADH: May increase therapeutic effects of Vasopressin. Specifically, the pressor and antidiuretic effects of vasopressin may be increased. *Risk C: Monitor*

Esketamine (Injection): Vasopressin may increase adverse/toxic effects of Esketamine (Injection). Specifically, the risk for elevated heart rate, hypertension, and arrhythmias may be increased. *Risk C: Monitor*

Etrasimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Fexinidazole: Bradycardia-Causing Agents may increase arrhythmogenic effects of Fexinidazole. *Risk X: Avoid*

Fingolimod: Bradycardia-Causing Agents may increase bradycardic effects of Fingolimod. Management: Consult with the prescriber of any bradycardia-causing agent to see if the agent could be switched to an agent that does not cause bradycardia prior to initiating fingolimod. If combined, perform continuous ECG monitoring after the first fingolimod dose. *Risk D: Consider Therapy Modification*

Indomethacin (Systemic): May increase therapeutic effects of Vasopressin. Specifically, vasopressin effects on cardiac index and systemic vascular resistance may be enhanced. *Risk C: Monitor*

Ivabradine: Bradycardia-Causing Agents may increase bradycardic effects of Ivabradine. *Risk C: Monitor*

Lacosamide: Bradycardia-Causing Agents may increase AV-blocking effects of Lacosamide. *Risk C: Monitor*

Landiolol: Bradycardia-Causing Agents may increase bradycardic effects of Landiolol. *Risk X: Avoid*

Mecamylamine: May increase therapeutic effects of Vasopressin. Specifically, the effect of vasopressin on mean arterial blood pressure may be increased. *Risk C: Monitor*

Midodrine: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Ozanimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Ponesimod: Bradycardia-Causing Agents may increase bradycardic effects of Ponesimod. Management: Avoid coadministration of ponesimod with drugs that may cause bradycardia when possible. If combined, monitor heart rate closely and consider obtaining a cardiology consult. Do not initiate ponesimod in patients on beta-blockers if HR is less than 55 bpm. *Risk D: Consider Therapy Modification*

Siponimod: Bradycardia-Causing Agents may increase bradycardic effects of Siponimod. Management: Avoid coadministration of siponimod with drugs that may cause bradycardia. If combined, consider obtaining a cardiology consult regarding patient monitoring. *Risk D: Consider Therapy Modification*

Pregnancy Considerations

Vasopressin may produce tonic uterine contractions. Due to pregnancy-induced physiologic changes, vasopressin clearance may be increased in the second and third trimesters; dose adjustment may be needed. Clearance returns to normal within 2 weeks postpartum.

Breastfeeding Considerations

It is not known if vasopressin is present in breast milk.

Monitoring Parameters

During therapy: Serum and urine sodium, urine-specific gravity, urine and serum osmolality; urine output, fluid input and output, BP, heart rate, digital/extremity infusion site for blanching/extravasation.

After discontinuation of therapy: Electrolytes, fluid status, and urine output (to assess for reversible arginine vasopressin disorders (formerly called diabetes insipidus).

Mechanism of Action

Vasopressin stimulates a family of arginine vasopressin (AVP) receptors, oxytocin receptors, and purinergic receptors (Russell 2011). Vasopressin, at therapeutic doses used for vasodilatory shock, stimulates the AVPR1a (or V1) receptor and increases systemic vascular resistance and mean arterial blood pressure; in response to these effects, a decrease in heart rate and cardiac output may be seen. When the AVPR2 (or V2) receptor is stimulated, cyclic adenosine monophosphate (cAMP) increases which in turn increases water permeability at the renal tubule resulting in decreased urine volume and increased osmolality. Vasopressin, at pressor doses, also causes smooth muscle contraction in the GI tract by stimulating muscular V1 receptors and release of prolactin and ACTH via AVPR1b (or V3) receptors.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action:

Antidiuretic: Peak effect: 1 to 2 hours (Murphy-Human 2010).

Vasopressor effect: IV: Rapid with peak effect occurring within 15 minutes of initiation of continuous IV infusion.

Duration: SUBQ: Antidiuretic: 2 to 8 hours; IV: Vasopressor effect: Within 20 minutes after IV infusion terminated.

Absorption: None from GI tract, destroyed by trypsin in GI tract, must be administered parenterally.

Distribution: V_d: 140 mL/kg.

Protein binding: None.

Metabolism: Hepatic, renal (inactive metabolites).

Half-life elimination: IV, SubQ: 10 to 20 minutes (apparent half-life: ≤10 minutes).

Excretion: SUBQ: Urine (5% as unchanged drug) after 4 hours; IV: Urine (~6% as unchanged drug).

Patient Counseling Points

(For additional information see ”Vasopressin: Patient drug information”)

What is this drug used for?

• It is used to treat low blood pressure.

• If you have been given this drug for some other reason, talk with your doctor about the benefits and risks. Talk with your doctor if you have questions or concerns about using this drug.

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• Low sodium levels like headache, trouble focusing, memory problems, feeling confused, weakness, seizures, or change in balance

• Any unexplained bruising or bleeding

• Severe stomach pain

• Chest pain or pressure

• Fast, slow, or abnormal heartbeat

• Shortness of breath, a big weight gain, or swelling in the arms or legs

• Not able to pass urine or change in how much urine is passed

• Change in skin color to black or purple

• Skin sores

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (QA) Qatar: Embesin | Vasogecta

REFERENCES

1. Alharfi IM, Stewart TC, Foster J, Morrison GC, Fraser DD. Central diabetes insipidus in pediatric severe traumatic brain injury. *Pediatr Crit Care Med*. 2013;14(2):203-209. [PubMed 23314181]
2. Ali KF, San Martin VT, El-Hage L, et al. A newly devised vasopressin drip protocol in the management of chronic central diabetes insipidus during facial transplantation surgery. *AACE Clin Case Rep*. 2018;4(6):e497-e500.
3. Alten JA, Borasino S, Toms R, Law MA, Moellinger A, Dabal RJ. Early initiation of arginine vasopressin infusion in neonates after complex cardiac surgery. *Pediatr Crit Care Med*. 2012;13(3):300-304. [PubMed 21926664]
4. American Society of Health-System Pharmacists (ASHP). Pediatric continuous infusion standards. https://www.ashp.org/-/media/assets/pharmacy-practice/s4s/docs/Pediatric-Infusion-Standards.ashx. Updated March 2024. Accessed April 23, 2024.
5. Bag R, O'Connor MF, Dalton A, Marklin GF. Management of the deceased organ tissue. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed April 10, 2024.
6. Bauer SR, Aloi JJ, Ahrens CL, Yeh JY, Culver DA, Reddy AJ. Discontinuation of vasopressin before norepinephrine increases the incidence of hypotension in patients recovering from septic shock: a retrospective cohort study. *J Crit Care*. 2010;25(2):362.e7-362.e11. doi:10.1016/j.jcrc.2009.10.005 [PubMed 19926252]
7. Bauer SR, Culver DA, Abraham S, et al. Detectability of vasopressin in continuous venovenous hemodialysis effluent of patients with vasodilatory shock treated with exogenous arginine vasopressin. *Pharmacotherapy*. 2011;31(9):857-862. doi:10.1592/phco.31.9.857 [PubMed 21923586]
8. Bauer SR, Sacha GL, Lam SW, et al. Hemodynamic response to vasopressin dosage of 0.03 units/min vs. 0.04 units/min in patients with septic shock. *J Intensive Care Med*. 2020:885066620977181. doi:10.1177/0885066620977181 [PubMed 33251906]
9. Bidegain M, Greenberg R, Simmons C, et al, "Vasopressin for Refractory Hypotension in Extremely Low Birth Weight Infants," *J Pediatr*, 2010, 157(3):502-4. [PubMed 20727442]
10. Bissell BD, Magee C, Moran P, Bastin MLT, Flannery AH. Hemodynamic instability secondary to vasopressin withdrawal in septic shock. *J Intensive Care Med*. 2019;34(9):761-765. doi:10.1177/0885066617716396 [PubMed 28750598]
11. Brierley J, Carcillo JA, Choong K. Clinical practice parameters for hemodynamic support of pediatric and neonatal septic shock: 2007 update from the American College of Critical Care Medicine. *Crit Care Med*. 2009;37(2):666-688. [PubMed 19325359]
12. Burton GL, Kaufman J, Goot BH, da Cruz EM. The use of arginine vasopressin in neonates following the Norwood procedure. *Cardiol Young*. 2011;21(5):536-544. [PubMed 21554826]
13. Callahan DS, Neville A, Bricker S, et al. The effect of arginine vasopressin on organ donor procurement and lung function. *J Surg Res*. 2014;186(1):452-457. doi:10.1016/j.jss.2013.09.028 [PubMed 24176209]
14. Chanson P, Jedynak CP, Dabrowski G, et al. Ultralow doses of vasopressin in the management of diabetes insipidus. *Crit Care Med*. 1987;15(1):44-46. doi:10.1097/00003246-198701000-00010 [PubMed 2431834]
15. Chojkier M, Groszmann RJ, Atterbury CE, et al. A Controlled Comparison of Continuous Intraarterial and Intravenous Infusions of Vasopressin in Hemorrhage From Esophageal Varices. *Gastroenterology*. 1979;77(3):540-546. [PubMed 313353]
16. Choong K and Kissoon N. Vasopressin in pediatric shock and cardiac arrest. *Pediatr Crit Care Med.* 2008;9(4):372-379. doi:10.1097/PCC.0b013e318172d7c8 [PubMed 18496412]

17. Choong K, Bohn D, Fraser DD, et al. Vasopressin in pediatric vasodilatory shock: a multicenter randomized controlled trial. *Am J Respir Crit Care Med.* 2009;180(7):632-639. doi:10.1164/rccm.200902-0221OC [PubMed 19608718]

18. Davalos MC, Barrett R, Seshadri S, et al. Hyponatremia during arginine vasopressin therapy in children following cardiac surgery. *Pediatr Crit Care Med*. 2013;14(3):290-297. [PubMed 23392370]

19. Dellinger RP, Levy MM, Rhodes A, et al. Surviving sepsis campaign: international guidelines for management of severe sepsis and septic shock: 2012. *Crit Care Med*. 2013;41(2):580-637. doi:10.1097/CCM.0b013e31827e83af [PubMed 23353941]

20. Denkler KA, Cohen BE. Reversal of dopamine extravasation injury with topical nitroglycerin ointment. *Plast Reconstr Surg*. 1989;84(5):811-813. [PubMed 2510208]

21. Duncan JM, Meaney P, Simpson P, et al. Vasopressin for in-hospital pediatric cardiac arrest: results from the American Heart Association National Registry of Cardiopulmonary Resuscitation. *Pediatr Crit Care Med.* 2009;10(2):191-955. doi:10.1097/PCC.0b013e31819a36f2 [PubMed 19188873]

22. Dünser MW, Mayr AJ, Tür A, et al. Ischemic skin lesions as a complication of continuous vasopressin infusion in catecholamine-resistant vasodilatory shock: incidence and risk factors. *Crit Care Med*. 2003;31(5):1394-1398. doi:10.1097/01.CCM.0000059722.94182.79 [PubMed 12771608]

23. Dunser MW, Mayr AJ, Ulmer H, et al, “The Effects of Vasopressin on Systemic Hemodynamics in Catecholamine-Resistant Septic and Postcardiotomy Shock: A Retrospective Analysis,” *Anest Analg*, 2001, 93(1):7-13.

24. Ettema EM, Heida J, Casteleijn NF, et al. The effect of renal function and hemodialysis treatment on plasma vasopressin and copeptin levels. *Kidney Int Rep*. 2017;2(3):410-419. doi:10.1016/j.ekir.2017.01.006 [PubMed 29142968]

25. Evans L, Rhodes A, Alhazzani W, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock 2021. *Crit Care Med.* 2021;49(11):e1063-e1143. doi:10.1097/CCM.0000000000005337 [PubMed 34605781]

26. Expert opinion. Senior Renal Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

27. Fleseriu M, Hashim IA, Karavitaki N, et al. Hormonal replacement in hypopituitarism in adults: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab*. 2016;101(11):3888-3921. doi:10.1210/jc.2016-2118 [PubMed 27736313]

28. Gal P, Reed M. Medications. In: Kliegman RM, Behrman RE, Jenson HB, et al, eds. *Nelson Textbook of Pediatrics*. 18th ed. Philadelphia, PA: Saunders Elsevier; 2007: 2955-2999.

29. Garcia-Tsao G, Sanyal AJ, Grace ND, Carey W; Practice Guidelines Committee of the American Association for the Study of Liver Diseases; Practice Parameters Committee of the American College of Gastroenterology. Prevention and management of gastroesophageal varices and variceal hemorrhage in cirrhosis [published correction appears in *Hepatology*. 2007;46(6):2052]. *Hepatology*. 2007;46(3):922-938. [PubMed 17879356]

30. Gazmuri RJ and Shakeri SA, “Low-Dose Vasopressin for Reversing Vasodilation During Septic Shock,” *Crit Care Med*, 2001, 29(3):673-5. [PubMed 11373445]

31. Hadaway L, ”Infiltration and Extravasation,” *Am J Nurs*, 2007, 107(8):64-72. [PubMed 17667395]

32. Hammond DA, McCain K, Painter JT, et al. Discontinuation of vasopressin before norepinephrine in the recovery phase of septic shock [published online ahead of print January 1, 2017]. *J Intensive Care Med*. doi: 10.1177/0885066617714209. [PubMed 28618919]

33. Hodges BM and Fraser G, ”Vasopressin for Vasodilatory Shock,” *Hosp Pharm*, 2002, 37(11):1149-57.

34. Hollenberg SM, Ahrens TS, Annane D, et al, “Practice Parameters for Hemodynamic Support of Sepsis in Adult Patients: 2004 Update,” *Crit Care Med*, 2004, 32(9):1928-48. [PubMed 15343024]

35. Holmes CL, Walley KR, Chittock DR, et al, “The Effects of Vasopressin on Hemodynamics and Renal Function in Severe Septic Shock: A Case Series,” *Inten Care Med*, 2001, 27(8):1416-21.

36. Ikegami H, Funato M, Tamai H, et al, ”Low-Dose Vasopressin Infusion Therapy for Refractory Hypotension in ELBW Infants,” *Pediatr Int*, 2010, 52(3):368-73. [PubMed 19793209]

37. Jeon K, Song JU, Chung CR, Yang JH, Suh GY. Incidence of hypotension according to the discontinuation order of vasopressors in the management of septic shock: a prospective randomized trial (DOVSS). *Crit Care*. 2018;22(1):131. doi:10.1186/s13054-018-2034-9 [PubMed 29784057]

38. Kahn JM, Kress JP, Hall JB. Skin Necrosis After Extravasation of Low-Dose Vasopressin Administered for Septic Shock. *Crit Care Med*. 2002;30(8):1899-1901. [PubMed 12163813]

39. Katz K, Lawler J, Wax J, et al. Vasopressin Pressor Effects in Critically Ill Children During Evaluation for Brain Death and Organ Recovery. *Resuscitation*. 2000;47(1):33-40. [PubMed 11004379]

40. Kill C, Wranze E, Wulf H. Successful Treatment of Severe Anaphylactic Shock With Vasopressin. *Int Arch Allergy Immunol*. 2004;134(3):260-261. [PubMed 15178897]

41. Kleinman ME, Chameides L, Schexnayder SM, et al. Part 14: Pediatric Advanced Life Support: 2010 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. *Circulation.* 2010;122(18)(suppl 3):S876-S908. doi:10.1161/CIRCULATIONAHA.110.971101 [PubMed 20956230]

42. Klinzing S, Simon M, Reinhart K, Bredle DL, Meier-Hellmann A. High-dose vasopressin is not superior to norepinephrine in septic shock. *Crit Care Med*. 2003;31(11):2646-2650. doi:10.1097/01.CCM.0000094260.05266.F4 [PubMed 14605537]

43. Koshman SL, Zed PJ, Abu-Laban RB. Vasopressin in Cardiac Arrest. *Ann Pharmacother*. 2005;39(10):1687-1692. [PubMed 16118265]

44. Kotloff RM, Blosser S, Fulda GJ, et al; Society of Critical Care Medicine/American College of Chest Physicians/Association of Organ Procurement Organizations Donor Management Task Force. Management of the potential organ donor in the ICU: Society of Critical Care Medicine/American College of Chest Physicians/Association of Organ Procurement Organizations consensus statement. *Crit Care Med*. 2015;43(6):1291-1325. doi:10.1097/CCM.0000000000000958 [PubMed 25978154]

45. Kristeller JL, Sterns RH. Transient diabetes insipidus after discontinuation of therapeutic vasopressin. *Pharmacotherapy*. 2004;24(4):541-545. [PubMed 15098812]

46. Landry DW, Levin HR, Gallant EM, et al. Vasopressin Deficiency Contributes to the Vasodilation of Septic Shock. *Circulation*. 1997;95(5):1122-1125. [PubMed 9054839]

47. Landry DW, Levin HR, Gallant EM, et al. Vasopressin Pressor Hypersensitivity in Vasodilatory Septic Shock. *Crit Care Med*. 1997;25(8):1279-1282. [PubMed 9267938]

48. Levitt MA, Fleischer AS, Meislin HW. Acute Post-traumatic Diabetes Insipidus: Treatment with Continuous Intravenous Vasopressin. *J Trauma*. 1984;24(6):532-535. [PubMed 6737531]

49. Levy MM, Evans LE, Rhodes A. The surviving sepsis campaign bundle: 2018 update. *Crit Care Med*. 2018;46(6):997-1000. doi:10.1097/CCM.0000000000003119 [PubMed 29767636]

50. Liedel JL, et al. The Successful Use of Vasopressin in Children With Hypotension. *Crit Care Med*. 2001;29(12):A63.

51. Lindeman RD, Lee TD Jr, Yiengst MJ, et al. Influence of Age, Renal Disease, Hypertension, Diuretics, and Calcium on the Antidiuretic Responses to Suboptimal Infusions of Vasopressin. *J Lab Clin Med*. 1966;68(2):206-223. [PubMed 5912596]

52. Malay MB, Ashton RC, Landry DW, et al. Low-Dose Vasopressin in the Treatment of Vasodilatory Septic Shock. *J Trauma*. 1999;47(4):699-705. [PubMed 10528604]

53. Manaker S, Teja B. Use of vasopressors and inotropes. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed November 8, 2021.

54. Mann K, Berg RA, and Nadkarni V. Beneficial effects of vasopressin in prolonged pediatric cardiac arrest: a case series. *Resuscitation.* 2002;52(2):149-156. [PubMed 11841882]

55. Mentzelopoulos SD, Malachias S, Chamos C, et al. Vasopressin, steroids, and epinephrine and neurologically favorable survival after in-hospital cardiac arrest: a randomized clinical trial. *JAMA*. 2013;310(3):270-279. [PubMed 23860985]

56. Mentzelopoulos SD, Zakynthinos SG, Tzoufi M, et al. Vasopressin, epinephrine, and corticosteroids for in-hospital cardiac arrest. *Arch Intern Med*. 2009;169(1):15-24. [PubMed 19139319]

57. Meyer S, Gortner L, McGuire W, et al. Vasopressin in catecholamine-refractory shock in children. *Anaesthesia.* 2008;63(3):228-234. doi:10.1111/j.1365-2044.2007.05317.x [PubMed 18081903]

58. Miller JH, Shock NW. Age Differences in the Renal Tubular Response to Antidiuretic Hormone. *J Gerontol*. 1953;8(4):446-450. [PubMed 13130853]

59. Mohamed A, Nasef N, Shah V, McNamara PJ. Vasopressin as a rescue therapy for refractory pulmonary hypertension in neonates: case series. *Pediatr Crit Care Med*. 2014;15(2):148-154. [PubMed 24141655]

60. Murphy-Human T, Diringer MN. Sodium disturbances commonly encountered in the neurologic intensive care unit. *J Pharm Pract*. 2010;23(5):470-482. doi:10.1177/0897190010372323 [PubMed 21507850]

61. Musallam N, Altshuler D, Merchan C, Zakhary B, Aberle C, Papadopoulos J. Evaluating vasopressor discontinuation strategies in patients with septic shock on concomitant norepinephrine and vasopressin infusions. *Ann Pharmacother*. 2018;52(8):733-739. doi:10.1177/1060028018765187 [PubMed 29560736]

62. Nakagawa TA. Pediatric Donor Management Guidelines. NATCO, The Organization for Transplant Professionals. 2008. *UpToDate®*. Basow DS, ed. Waltham, MA: *UpToDate®*; 2011. http://www.uptodate.com

63. Ong J, Van Gerpen R. Recommendations for management of noncytotoxic vesicant extravasations. *J Infus Nurs*. 2020;43(6):319-343. doi:10.1097/NAN.0000000000000392 [PubMed 33141794]

64. Perman SM, Elmer J, Maciel CB, et al. 2023 American Heart Association focused update on adult advanced cardiovascular life support: an update to the American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation*. 2024;149(5):e254-e273. doi:10.1161/CIR.0000000000001194 [PubMed 38108133]

65. Plurad DS, Bricker S, Neville A, Bongard F, Putnam B. Arginine vasopressin significantly increases the rate of successful organ procurement in potential donors. *Am J Surg*. 2012;204(6):856-860; discussion 860-861. doi:10.1016/j.amjsurg.2012.05.011 [PubMed 23116641]

66. Prescott HC, Antonelli M, Alhazzani W, et al. Surviving sepsis campaign: international guidelines for management of sepsis and septic shock 2026. *Crit Care Med*. 2026;54(4):725-812. doi:10.1097/CCM.0000000000007075 [PubMed 41869847]

67. Prete A, Corsello SM, Salvatori R. Current best practice in the management of patients after pituitary surgery. *Ther Adv Endocrinol Metab*. 2017;8(3):33-48. doi:10.1177/2042018816687240 [PubMed 28377801]

68. Rana H, Ferguson N, Dicpinigaitis PV. Diabetes insipidus after discontinuation of vasopressin infusion for septic shock. *J Clin Pharm Ther*. 2018;43(2):287-290. doi:10.1111/jcpt.12627 [PubMed 28895166]

69. Reid IA. Role of Vasopressin Deficiency in the Vasodilation of Septic Shock. *Circulation*. 1997;95(5):1108-1110. [PubMed 9054835]

70. Rhodes A, Evans LE, Alhazzani W, et al. Surviving Sepsis Campaign: International guidelines for management of sepsis and septic shock: 2016. *Intensive Care Med*. 2017;43(3):304-377. doi:10.1007/s00134-017-4683-6 [PubMed 28101605]

71. Rios DR, Kaiser JR. Vasopressin versus dopamine for treatment of hypotension in extremely low birth weight infants: a randomized, blinded pilot study. *J Pediatr*. 2015;166(4):850-855. [PubMed 25641242]

72. Rios DR, Moffett BS, Kaiser JR. Trends in pharmacotherapy for neonatal hypotension. *J Pediatr*. 2014;165(4):697-701. [PubMed 25039051]

73. Rosendale JD, Kauffman HM, McBride MA, et al. Aggressive Pharmacologic Donor Management Results in More Transplanted Organs. *Transplantation*. 2003a;75(4):482-487. [PubMed 12605114]

74. Rosendale JD, Kauffman HM, McBride MA, et al. Hormonal resuscitation yields more transplanted hearts, with improved early function. *Transplantation*. 2003b;75(8):1336-1341. [PubMed 12717226]

75. Rosenzweig EB, Starc TJ, Chen JM, et al, ”Intravenous Arginine-Vasopressin in Children With Vasodilatory Shock After Cardiac Surgery,” *Circulation*, 1999, 100(19)(suppl):II182-6. [PubMed 10567301]

76. Rozenfeld V, Cheng JW. The Role of Vasopressin in the Treatment of Vasodilation in Shock States. *Ann Pharmacother*. 2000;34(2):250-254. [PubMed 10676834]

77. Russell JA. Bench-to-bedside review: Vasopressin in the management of septic shock. *Crit Care*. 2011;15(4):226. [PubMed 21892977]

78. Russell JA, Walley KR, Singer J, et al; VASST Investigators. VVasopressin versus norepinephrine infusion in patients with septic shock. *N Engl J Med*. 2008;358(9):877-887. doi:10.1056/NEJMoa067373 [PubMed 18305265]

79. Russell JA, Gordon AC, Williams MD, Boyd JH, Walley KR, Kissoon N. Vasopressor therapy in the intensive care unit. *Semin Respir Crit Care Med*. 2021;42(1):59-77. doi:10.1055/s-0040-1710320 [PubMed 32820475]

80. Sacha GL, Lam SW, Duggal A, Torbic H, Reddy AJ, Bauer SR. Hypotension risk based on vasoactive agent discontinuation order in patients in the recovery phase of septic shock. *Pharmacotherapy*. 2018;38(3):319-326. doi:10.1002/phar.2082 [PubMed 29328496]

81. Schummer C, Wirsing M, Schummer W. The Pivotal Role of Vasopressin in Refractory Anaphylactic Shock. *Anesth Analg*. 2008;107(2):620-624. [PubMed 18633042]

82. Smolders EJ, Benoist GE, Smit CCH, Ter Horst P. An update on extravasation: basic knowledge for clinical pharmacists. *Eur J Hosp Pharm*. 2020;28(3):165-167. doi:10.1136/ejhpharm-2019-002152 [PubMed 32341081]

83. Sperling MA, ed. *Pediatric Endocrinology*. 4th ed. Canada: Elsevier; 2014.

84. Stefanos SS, Kiser TH, MacLaren R, Mueller SW, Reynolds PM. Management of noncytotoxic extravasation injuries: a focused update on medications, treatment strategies, and peripheral administration of vasopressors and hypertonic saline. *Pharmacotherapy*. 2023;43(4):321-337. doi:10.1002/phar.2794 [PubMed 36938775]

85. Stiell IG, Hebert PC, Wells GA, et al. Vasopressin Versus Epinephrine for In hospital Cardiac Arrest: A Randomised Controlled Trial. *Lancet*. 2001;358(9276):105-109. [PubMed 11463411]

86. Traut U, Brügger L, Kunz R, et al. Systemic prokinetic pharmacologic treatment for postoperative adynamic ileus following abdominal surgery in adults. *Cochrane Database Syst Rev*. 2008;(1):CD004930. doi:10.1002/14651858.CD004930.pub3 [PubMed 18254064]

87. Tsuneyoshi I, Yamada H, Kakihana Y, et al. Hemodynamic and Metabolic Effects of Low-Dose Vasopressin Infusions in Vasodilatory Septic Shock. *Crit Care Med*. 2001;29(3):487-493. [PubMed 11373409]

88. Tuggle DW, Bennett KG, Scott J, Tunell WP. Intravenous vasopressin and gastrointestinal hemorrhage in children, *J Pediatr Surg*. 1988;23(7):627-629. [PubMed 3144593]

89. UNOS. Critical Pathway for Donation After Cardiac Death.

90. van der Zee S, Thompson A, Zimmerman R, et al. Vasopressin administration facilitates fluid removal during hemodialysis. *Kidney Int*. 2007;71(4):318-324. doi:10.1038/sj.ki.5001885 [PubMed 17003815]

91. Van Matre ET, Rice PJ, Wempe MF, et al. Extended stability of vasopressin injection in polyvinyl chloride bags and polypropylene syringes and its impact on critically ill patient care and medication waste. *Hosp Pharm*. 2023;58(2):205-211. doi:10.1177/00185787221130229 [PubMed 36890958]

92. Vasopressin [product monograph]. Toronto, Ontario, Canada: Fresenius Kabi Canada Ltd; March 2022.

93. Vasopressin in sodium chloride injection bags [prescribing information]. Deerfield, IL: Baxter Healthcare Corporation; February 2024.

94. Vasopressin in sodium chloride injection ready to use vials [prescribing information]. Rosemont, IL: Long Grove Pharmaceuticals; April 2024.

95. Vasopressin vial [prescribing information]. Shirley, NY: American Regent Inc; August 2020.

96. Vasostrict (vasopressin) [prescribing information]. Malvern, PA: Endo USA; March 2025.

97. Wigginton JG, Agarwal S, Bartos JA, et al. Part 9: adult advanced life support: 2025 American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation*. 2025;152(16 suppl 2):S538-S577. doi:10.1161/CIR.0000000000001376 [PubMed 41122884]

98. Williams SR, Denault AY, Pellerin M, et al. Vasopressin for Treatment of Shock Following Aprotinin Administration. *Can J Anaesth*. 2004;51(2):169-172. [PubMed 14766695]

99. Wise-Faberowski L, Soriano SG, Ferrari L, et al. Perioperative management of diabetes insipidus in children. *J Neurosurg Anesthesiol*. 2004;16(3):220-225. [PubMed 15211159]

100. Wood KE, Becker BN, McCartney JG, et al. Care of the Potential Organ Donor. *N Engl J Med*. 2004;351(26):2730-2739. [PubMed 15616207]

101. Wong AF, McCulloch LM, Sola A. Treatment of peripheral tissue ischemia with topical nitroglycerin ointment in neonates. *J Pediatr*. 1992;121(6):980-983. [PubMed 1447671]

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